

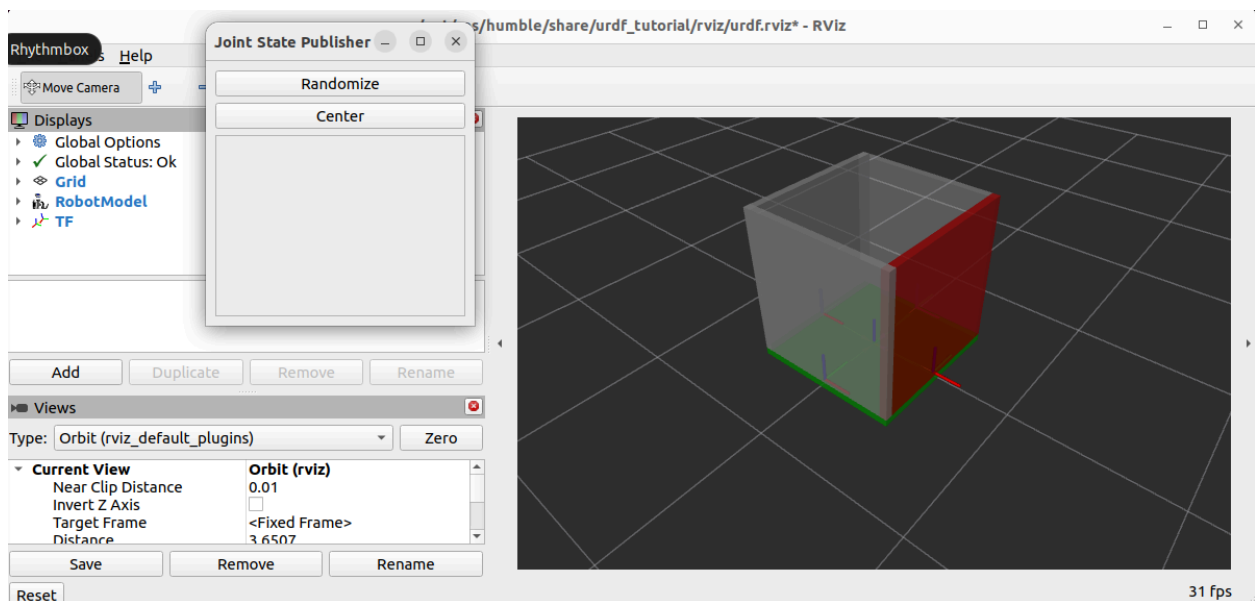
Prelab

- Provide a reflection on the assessing and expanding background knowledge sections.

The assessing background knowledge section helped reinforce the XML rules needed for URDF, especially that tags must close properly, nesting must be correct, and tag names cannot contain spaces. The expanding background knowledge section helped connect those XML rules to robot modeling by showing how links, joints, materials, and origins work together to create and position robot parts in RViz.

- Add at least one material to the box created in the activity.

- Provide the URDF file and a screenshot in Rviz (with joint state publisher visible) of the completed box with your modification.



box.urdf below

```
<?xml version="1.0"?>
<robot name="box">

  <material name="green">
    <color rgba="0 0.8 0 1"/>
  </material>

  <material name="gray">
    <color rgba="0.7 0.7 0.7 1"/>
  </material>
```

```
<material name="red">
  <color rgba="0.8 0 0 1"/>
</material>
```

```
<link name="base_link">
  <visual>
    <geometry>
      <box size="1 1 0.05"/>
    </geometry>
    <material name="green"/>
  </visual>
</link>
```

```
<link name="back_link">
  <visual>
    <geometry>
      <box size="1 1 0.05"/>
    </geometry>
    <origin xyz="0 0 0.5" rpy="1.5708 0 1.5708"/>
    <material name="gray"/>
  </visual>
</link>
```

```
<joint name="back_to_bottom" type="fixed">
  <parent link="base_link"/>
  <child link="back_link"/>
  <origin xyz="-0.475 0 0.025" rpy="0 0 0"/>
</joint>
```

```
<link name="front_link">
  <visual>
    <geometry>
      <box size="1 1 0.05"/>
    </geometry>
    <origin xyz="0 0 0.5" rpy="1.5708 0 1.5708"/>
    <material name="red"/>
  </visual>
</link>
```

```
<joint name="front_to_bottom" type="fixed">
  <parent link="base_link"/>
  <child link="front_link"/>
  <origin xyz="0.475 0 0.025" rpy="0 0 0"/>
</joint>
```

```
<link name="left_link">
  <visual>
    <geometry>
```

```
<box size="1 1 0.05"/>
</geometry>
<origin xyz="0 0 0.5" rpy="1.5708 0 0"/>
<material name="gray"/>
</visual>
</link>

<joint name="left_to_bottom" type="fixed">
  <parent link="base_link"/>
  <child link="left_link"/>
  <origin xyz="0 -0.475 0.025" rpy="0 0 0"/>
</joint>

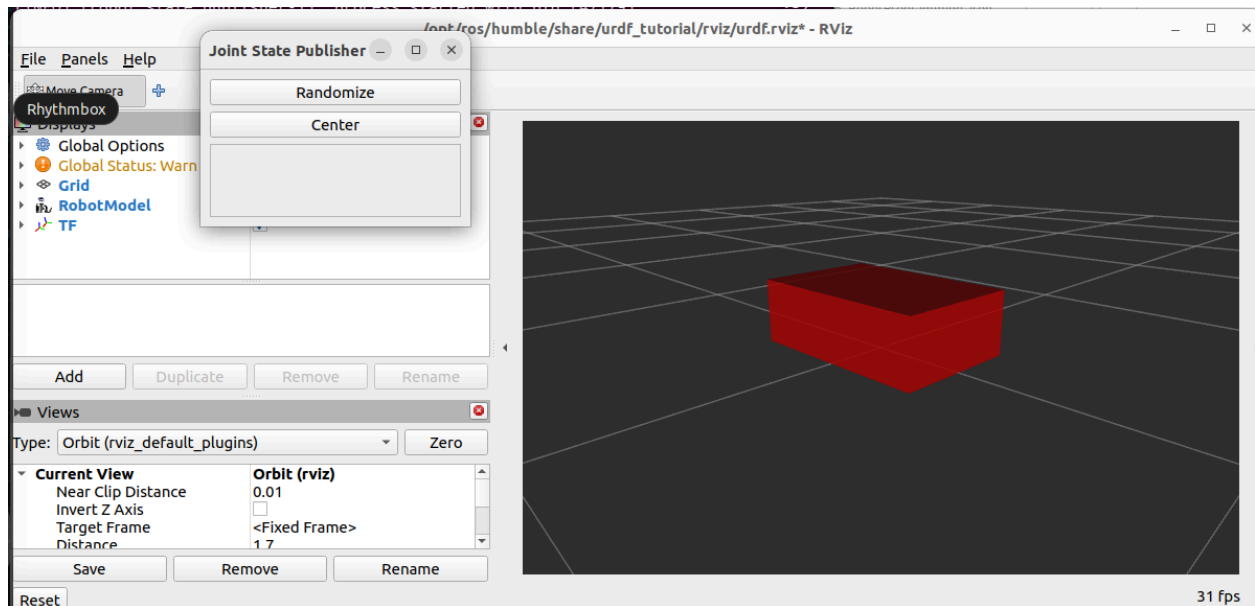
<link name="right_link">
  <visual>
    <geometry>
      <box size="1 1 0.05"/>
    </geometry>
    <origin xyz="0 0 0.5" rpy="1.5708 0 0"/>
    <material name="gray"/>
  </visual>
</link>

<joint name="right_to_bottom" type="fixed">
  <parent link="base_link"/>
  <child link="right_link"/>
  <origin xyz="0 0.475 0.025" rpy="0 0 0"/>
</joint>

</robot>
```

Lab 6 Deliverables

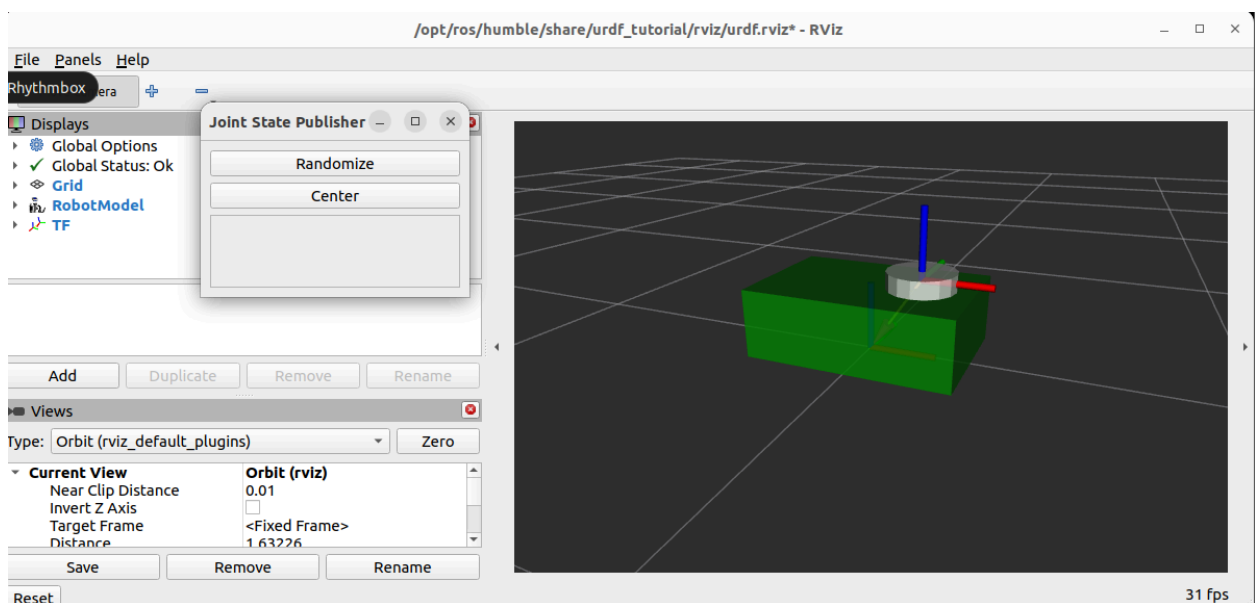
- Visual for Task 1 step 8



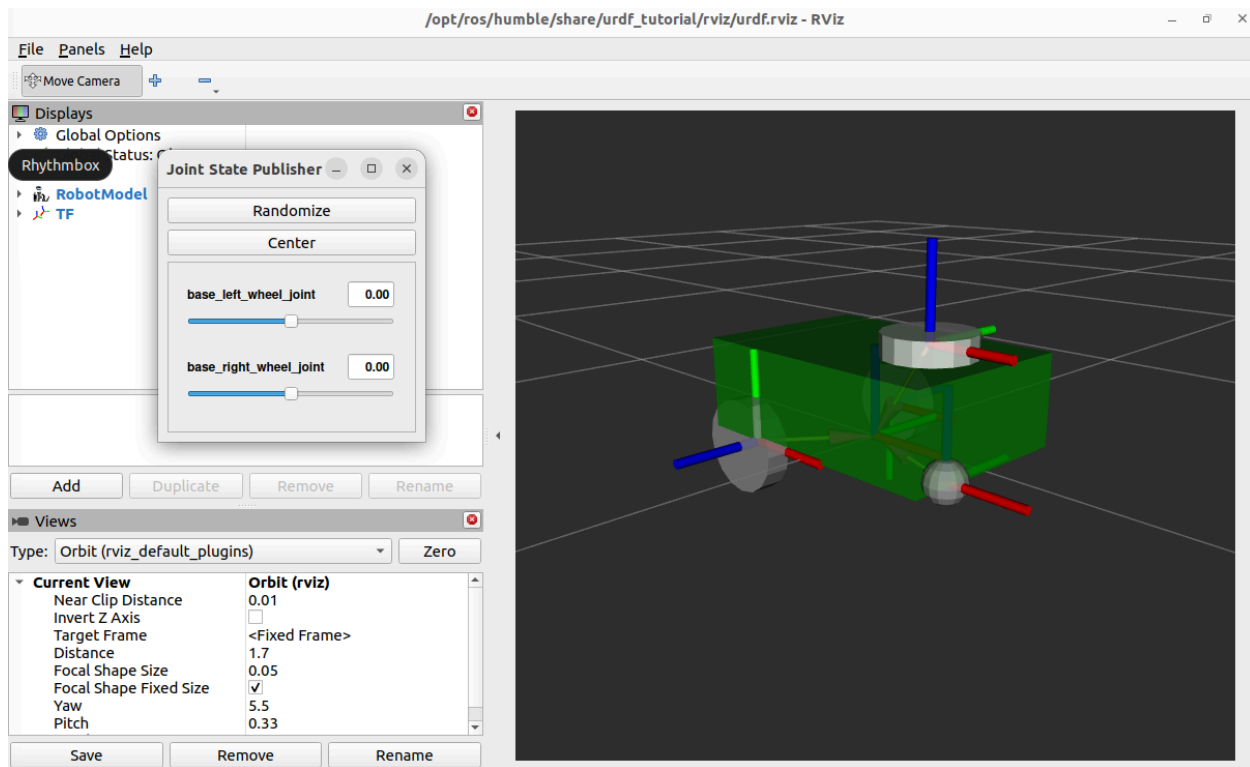
REQUIRED Question: What happens if you try to run the demo now? Why?

The robot will not be described properly because the new link is not connected to the existing robot structure. Every added link must be connected by a joint.

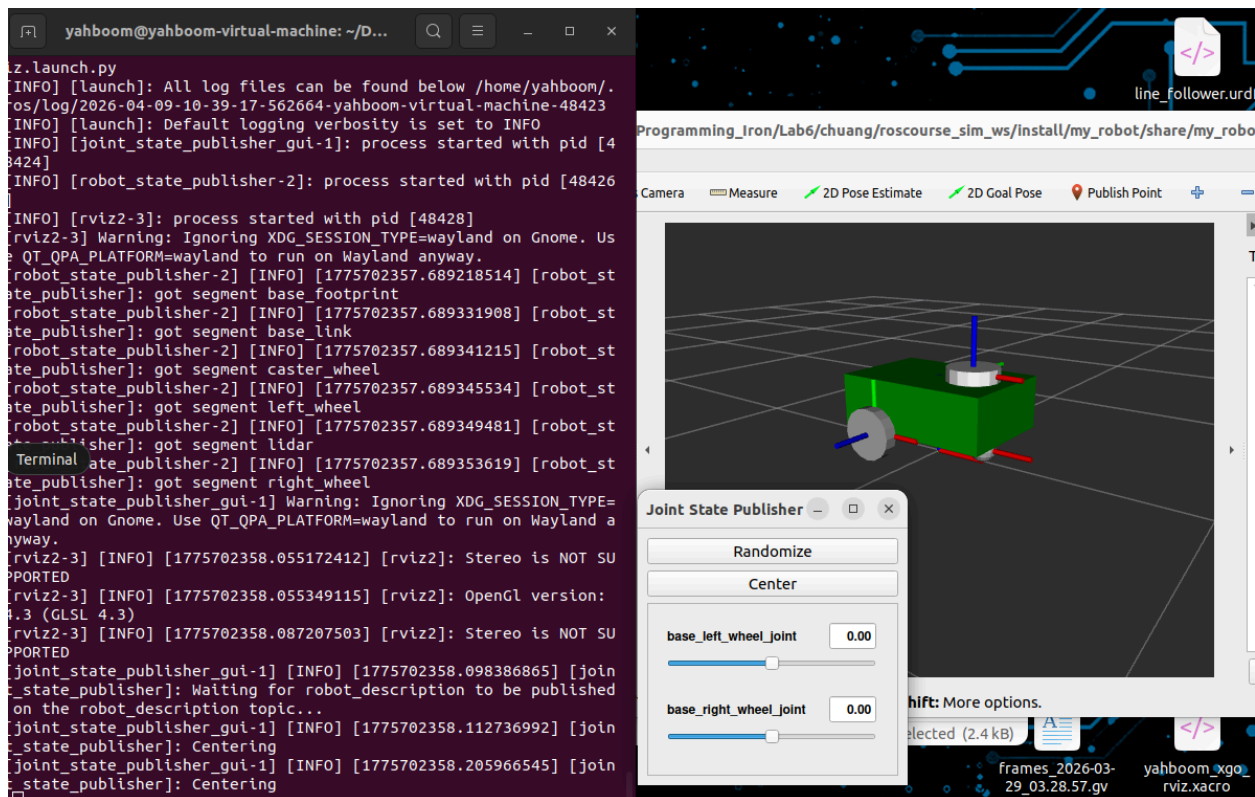
- Visual for Task 1 step 18



- Visual for Task 1 step 36



- Visual for Task 2 step 6 with launch terminal command visible



REQUIRED Question: What are some of the nodes in the launch file and what do they do?

joint_state_publisher publishes joint states when GUI mode is off

joint_state_publisher_gui publishes joint states using sliders when GUI mode is on

robot_state_publisher publishes the robot transforms and robot description from the URDF

rviz2 launches RViz and loads the configured display settings

- Visual for Task 3 step 3 (whichever version you used)

No webcam option

